

Danbee Kim, PhD

Educator | Neuroscientist | Performance Artist



SUMMARY

15+ years of experience using live performance and movement games **to teach** neuroanatomy, physiology, and systems biology **across university, industry, and community settings**. Specialized in translating complex concepts into accessible, engaging experiences for diverse learners. Extensive experience developing curricula, mentoring students, and delivering high-quality instruction in traditional and project-based environments. Demonstrated commitment to inclusive, curiosity-driven education.

EDUCATION

2013 to 2020
Lisboa, Portugal

PhD in Neuroscience
*Champalimaud Centre for the Unknown,
International Neuroscience Doctoral Programme*

2005 to 2009
Cambridge, USA

BSc in Brain and Cognitive Sciences (Course 9)
Massachusetts Institute of Technology

TEACHING EXPERIENCE

Lead Instructor

Matter Neuroscience | London, UK | 2023-2024

- Designed and delivered neuroscience and physiology curriculum for adult learners, interdisciplinary teams, and public audiences.
- Taught foundational concepts in neuroanatomy, neural signaling, sensory systems, and physiological regulation.
- Created **multimedia teaching materials and assessment tools for diverse learning styles**.

Course Co-director

CAJAL Advanced Neuroscience Training Programme | Multiple Locations | 2022-2024

- Designed and delivered "Modern Approaches to Behavioural Analysis", an annual one-week intensive training course for undergraduate, graduate, and post-doctoral researchers.
- Taught conceptual frameworks for studying biological behavior, fundamental concepts in nervous system anatomy, and experimental design principles for behavioral neuroscience.
- Organized online guest lecturers, local teaching assistants, and a course Discord server to provide further resources and 24/7 support to students joining the course from locations around the world.

Co-founder, Curriculum Developer, Instructor

Neuronautas, Academia Gulbenkian do Conhecimento | Lisboa, Portugal | 2018-2023

- Developed and taught a **two-week project-based curriculum** in neurobiology, physiology, philosophy of science, and robotics **for high-school and early-college students**.
- **Created hands-on modules** covering organ systems, sensory physiology, motor control, and homeostasis.
- Managed teaching assistants and **mentored students through independent group projects**.

Theory Tutor, Masters Thesis Advisor

Bartlett School of Architecture, University College London | London, UK | 2020-2022

- Taught biological principles -- including human physiology, perception, and systems neuroscience -- to graduate students in the **Design for Performance and Interaction MArch** course.
- Supervised thesis research, provided structured feedback, and fostered scientific communication skills.

Teacher, Staff

Appalachian Institute for Creative Learning | Mars Hill, North Carolina, USA | 2010-present

- Taught **interdisciplinary courses** including introductory biology, neuroanatomy, and organismal physiology in **mixed-age classrooms** at a summer camp for rising **3rd-12th grade students**.
- **Designed experiential learning activities** emphasizing scientific reasoning and systems thinking.

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RESEARCH EXPERIENCE

Research & Outreach Scientist

NeuroGEARS, Ltd | London, UK | 2021-2023

- Conducted research on **neural activity in naturalistic settings**; developed **tools for behavioral and physiological measurement**.
- Communicated scientific findings to interdisciplinary collaborators and public audiences.

PhD Researcher

Champalimaud Centre for the Unknown | Lisboa, Portugal | 2012-2020

Marine Biological Laboratory | Woods Hole, Massachusetts, USA | 2014-2015

Sainsbury Wellcome Centre for Neural Circuits and Behaviour | London, UK | 2017-2020

- Developed theoretical and methodological framework for **field neuroscience**, integrating **behavior, physiology, and environmental context**.
- Benchmarked eye-tracking as a non-invasive measure of nervous system activity in complex environments.
- Designed and published a non-invasive technique for simultaneously studying single-cell neural activity and whole organism behavior.

SELECTED PUBLICATIONS

(Abridged for relevance; full list available upon request)

- Kim, D. & Xiao, X. (2023). "**A Solarpunk Nervous System**". *Vector*, 297, 141-145.
- Kim, D. (2022) "**Are cephalopods making a comeback in neuroscience?**". *BioTechniques*.
- Kim, D. & Kampff, A.R. (2020) "**Neuroscience Does Design: What the Brain's Architecture Can Teach Architects**." *Architectural Design* 90.6: 94-99.
- Kim, D. (2020) **On the aims and methods of field neuroscience**. PhD dissertation. Universidade Nova de Lisboa. DOI: 10.13140/RG.2.2.35235.11042
- Kim, D. (2020) **The First VIRS**. London, UK: Danbee Kim.

RELEVANT ARTS EXPERIENCE

- Self-published and crowdfunded an **original YA sci-fi graphic novel** based on my PhD research, directed an **interactive aquarium installation** that engaged over 24,000 visitors, and co-wrote and choreographed an **original musical** produced by the MIT Musical Theater Guild.
- Developed neuroscience primer connecting **anatomy and physiology of nervous systems** to **social structures and culture**, adapted for learners ranging from 8 year olds to university researchers.
- PhD research required deep understanding of **behavioral observation, storytelling, and the impacts of social dynamics** on nervous system anatomy and physiology.
- Experience grounding interdisciplinary debates in **historical, political, and cultural contexts**.
- Strong background in **philosophy of science, speculative fiction, and live performance**.

CURRICULUM DEVELOPMENT & EDUCATIONAL LEADERSHIP

- Led strategic planning and content development for educational programs at Matter Neuroscience.
- Designed **multi-week curricula, lab-free physiology activities, and interactive learning modules** in settings such as aquariums, museums, and community centers.
- Created online learning environments, custom websites, and digital teaching tools that provided access to remote learners and supported organizational resilience during COVID-19 lockdowns.
- Launched a Junior TA program to **nurture student leadership** at Neuronautas.
- Managed teams of volunteer teaching assistants and **cultivated a collective philosophy of compassionate, multi-cultural, and liberatory education**.
- Participate as an alumni mentor, academic advisor, and life-long learner in intergenerational communities that emphasize **free-choice, project-based, and site-specific curricula**.

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TECHNICAL & TEACHING SKILLS

Live Performance

- **Musician:** vocals, violin, ukelele, acoustic double bass
- **Writer:** speculative fiction, theater scripts, graphic novel scripts, scientific writing, lyricist
- **Theater-maker:** stage acting & directing, vocal directing, choreography, pit orchestra musician, costume design, prop design, backstage crew

Digital Tools

- Python (numpy, scipy, OpenCV, matplotlib), MATLAB, R
- Bonsai-Rx, Arduino
- HTML/CSS/JS, Markdown, LaTeX
- Illustrator, Photoshop, Premiere Pro

Pedagogical Skills

- Curriculum design
- Project-based learning
- Inclusive, multi-cultural, and multi-modal teaching practices
- Assessment design
- Science communication

ADDITIONAL CREATIVE & COMMUNITY WORK

2024 to 2025 Walthamstow, UK	Co-director, Events Manager, Chef <i>Pulse & Pickle</i> CIC, a community cultural center and plant-based restaurant
2018 to 2020 London, UK	Author, Illustrator, Project Manager <i>The First VIRS</i> , a YA sci-fi graphic novel integrating neuroscience concepts
2017 to 2020 Brighton, UK	Scientific & Artistic Director <i>Surprising Minds</i> , an interactive aquarium exhibit engaging 24,000+ visitors
2005 to present Various Locations	Experienced live performer (theater, music, circus arts) with strong public-facing communication skills

SELECTED FUNDRAISING AWARDS & EXPERIENCE

- 2019 | Kickstarter | 155 backers, 145% funding for *The First VIRS*
- 2019 | Sainsbury Wellcome Centre for Neural Circuits and Behaviour | Public Engagement Fund
- 2018 | University College London | Train and Engage Fund
- 2018 | Fundação Calouste Gulbenkian | Academias Gulbenkian do Conhecimento (Neuronautas)

REFERENCES

Available upon request.